

Anu Ouseph

Working Student Software Developer

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PROFILE

Full-stack developer with with 2+ years of experience building real-time, interactive web applications. Strong focus on component architecture, WebSocket-driven interfaces, and visual/3D rendering using Three.js and GSAP. Comfortable working across the stack, Currently pursuing an M.Sc. in Research in Media Engineering.

PROFESSIONAL EXPERIENCE

Product Engineer – Hoomans.co (India)

06/2024 – 09/2025

- Built and maintained [Lyst.so](#), a real-time digital signage platform running on 500+ distributed displays and implemented WebSocket sync with reconnect logic and concurrent device state management (Vue.js, WebSockets, Go, PostgreSQL)
- Contributed frontend features to [TinkerHub.org](#), a tech community platform with 10K+ members, including TypeScript modules for user management, resource tracking, and event systems (Nuxt.js, TypeScript, GSAP)
- Built a custom interactive D3.js timeline visualization with scroll-driven animations and dynamic positioning for project timelines across multiple years (Vue.js, GSAP, D3.js) for [Hoomans.co](#)

PROJECTS

[MIRa – Music Information Retrieval Analysis](#)

Next.js · FastAPI · Python · Hugging face spaces · Vercel

- Built a full-stack music analysis and comparison tool where users upload audio files and get back tempo, key, genre, mood, and instrument features in real time
- Designed a RESTful FastAPI backend deployed on Hugging Face Spaces (Docker) with a Next.js frontend
- Implemented music similarity comparison using weighted cosine similarity over MFCC, chroma, spectral centroid, and tempo feature vectors

[Spatial – 3D Furniture Configurator](#)

React.js · Three.js · React three fiber · WebXR · Tailwind CSS

- Browser-based 3D furniture configurator with real GLTF models, real-time material customization, and WebXR AR - users can place furniture in their actual room via phone camera
- Implemented scene graph traversal and material cloning for per-part customization on multi-mesh models
- Responsive UI with desktop side panel and mobile bottom sheet, and AR-ready architecture built to real-world scale

Portfolio

Vue.js · Three.js · GLSL

- Interactive portfolio with custom GLSL shaders for mouse-driven image distortion and chromatic aberration effects
- Scroll-driven animations using GSAP ScrollTrigger with optimized rendering for smooth performance

EDUCATION

M.Sc. Research in Media Engineering

10/2025 – Present

Technische Universität Ilmenau, Germany

B.Tech. Electronics Communication Engineering

08/2018 – 07/2022

Govt. College of Engineering, Kannur, India

Full-Stack Developer Program – Brototype

11/2022 – 09/2023

Intensive project-based training: JavaScript, Vue.js, React, Node.js, MongoDB, PostgreSQL

TECHNICAL SKILLS

Programming Languages: JavaScript, TypeScript, Golang, Python

Frontend: React.js, Vue.js, Nuxt.js, Next.js, Three.js, GSAP, Tailwind CSS, Redux, Pinia

Backend: Node.js, Go, FastAPI, WebSockets, REST APIs, PostgreSQL, MongoDB, Supabase

Tools: Git, Docker, AWS (Lightsail, S3), Vercel, Hugging Face Spaces, CI/CD

Languages: English C1 (fluent) · German A2 (learning) · Malayalam (native)